

Han Liu

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Education

Ph.D. in Mechanical Engineering	Peking University , Fluid Mechanics	Beijing, China
	Thesis: Characteristics and action mechanism of the structures in ventilated supercavity	Sept 2012 – June 2018
B.S.	Peking University , Theoretical and Applied Mechanics	Beijing, China
	Thesis: Numerical investigation on heavy particle pair dispersion in the inertial range of two dimensional turbulence	Sept 2008 – June 2012
Certificate of Quantum Excellence	IBM Qiskit Global Summer School on Quantum Machine Learning , Quantum Computing and Quantum Machine Learning	2021
	Grade: 100/100	

Experience

Claros Technologies , CFD Consultant – Digital Twin Platform	Jan 2025 – present
- Built end-to-end CFD digital twin for PFAS chemical reactor with parametric geometry, simulation, data generation, and optimization - Integrated chemical kinetics into the solver and developed optimization algorithms for reactive flow design	
St. Anthony Falls Laboratory, University of Minnesota , Researcher	Minneapolis, MN Jan 2021 – present
- Built an incompressible CFD solver with AMR, coupled level-set/VOF interface tracking, and cut-cell embedded boundaries on AMReX (26K lines of C++, 12 modules, 467 unit tests, 7 validated benchmarks) - Ported lab's Fortran HPC code to C++/CUDA multi-GPU; breaking wave case (2048x2048x1024): 1 month on 2048 CPUs to 2.5 days on 64 A100s to under 5 hours with mixed-precision Poisson solver; 40% memory reduction - Developed hybrid quantum-classical framework (variational quantum circuits + physics-informed neural networks) for incompressible turbulence simulation - Applied quantum tensor networks (MPS/MPO) for data compression and acceleration in Navier-Stokes simulations	
St. Anthony Falls Laboratory, University of Minnesota , Post-Doctoral Associate	Minneapolis, MN Sept 2018 – Jan 2021
- Developed coupled level-set/VOF interface tracking for two-phase and three-phase flows at high density ratios (>1000:1); applied to oil spill simulation in turbulent breaking waves (JCP 2023) - Simulated COVID-19 airborne transmission in indoor environments; Featured Article in Physics of Fluids, covered by The Guardian - Researched wind-wave-bathymetry interactions near the surf zone for amphibious vehicle performance	

- Optimized hydro-turbine geometries using immersed boundary methods and large-eddy simulation

Dynamic Dimension Technologies (DDT), LLC, CFD Consultant – Ocean Wave Digital Twin

Jan 2023 – Sept 2023

- Built ocean wave modeling platform (shoaling, bathymetry, vehicle dynamics) with synthetic data generation for surrogate models

University of Minnesota, Visiting Scholar

Sept 2015 – Aug 2017

- Developed in-house code for simulating multiphase turbulent flow with VOF, level set, and immersed boundary methods
- Researched turbulent cavitating flows interacting with ocean waves

Peking University, Research Assistant

Beijing, China

Sept 2012 – June 2018

- Built a compressible high-Mach-number multiphase solver from scratch (Fortran + MPI); applied to shock-multiphase-flow interactions for inertial confinement fusion (Physical Review E)
- Developed coupled level-set/VOF interface tracking in an incompressible flow solver; studied underwater cavitation drag reduction (Journal of Fluid Mechanics)
- Investigated Richtmyer-Meshkov instability and turbulent mixing

Research Interests

- **Computational Fluid Dynamics (CFD)**: High-fidelity numerical simulation (DNS/LES), level set and volume of fluid methods, adaptive mesh refinement (AMR), immersed boundary and cut-cell methods
- **Multiphase Flow**: Air-water-oil systems, ventilated cavitation, bubble dynamics and tracking, high-density-ratio interface tracking
- **Turbulence**: Compressible turbulence, turbulent mixing, Richtmyer-Meshkov instability
- **High-Performance Computing**: Multi-GPU parallel computing (C++/CUDA), mixed-precision solvers, large-scale simulation optimization
- **Applied CFD**: COVID-19 airborne transmission, oil spill dynamics, hydrokinetic turbine design, aquatic canopy flows, digital twin platforms
- **Quantum-Inspired Computing**: Tensor network methods for Navier-Stokes equations, variational quantum algorithms for fluid mechanics, matrix product state encoding

Skills

HPC: Fortran, C++, CUDA, MPI, Python, Git, Slurm, PBS, HDF5, PETSc, hypre

Multiphase CFD: High-order schemes, interface tracking (level set, VOF, CLSVOF), Basilisk, AMReX, DNS/LES/RANS, turbulence modeling, immersed boundary method, shock capturing, Poisson solver, fluid-structure interaction, finite difference/volume, OpenFOAM, ANSYS

AI/ML: Physics-informed neural networks (PINNs), LLM-augmented development, multi-agent orchestration

Quantum Computing: Qiskit, Yao.jl, variational quantum circuits, quantum-PINNs, quantum tensor networks

Selected Publications

Quantum-inspired tensor network representations of the shallow water equation

Liu, H., Li, T., Shen, L.

Under revision

Simulation-based study of low Reynolds number flow around a ventilated cavity

2023

Liu, H., Xiao, Z., Shen, L.

Journal of Fluid Mechanics

A consistent adaptive level set framework for incompressible two-phase flows with high density ratios and high Reynolds numbers	2023
Zeng, Y., Liu, H. , et al. Journal of Computational Physics	
Simulation-based study of turbulent aquatic canopy flows with flexible stems	2022
He, S., Liu, H. , Shen, L. Journal of Fluid Mechanics	
Large-eddy simulation and co-design strategy for a drag-type vertical axis hydrokinetic turbine	2022
Gao, J., Liu, H. , Lee, J., Zheng, Y., Guala, M., Shen, L. Renewable Energy	
Refined modeling for single-mode cylindrical Richtmyer-Meshkov instability	2021
Wu, J., Liu, H. , Xiao, Z. Journal of Fluid Mechanics	
A robust and accurate technique for Lagrangian tracking of bubbles and detecting fragmentation and coalescence	2021
Gao, Q., Deane, G.B., Liu, H. , Shen, L. International Journal of Multiphase Flow	
Numerical investigation of ventilated cavitating flow in the wake of a circular cylinder	2021
Wang, Z., Liu, H. , Gao, Q., et al. Physical Review Fluids	
Simulation-based study of COVID-19 outbreak associated with air-conditioning in a restaurant [Featured Article]	2021
Liu, H. , He, S., Shen, L., Hong, J. Physics of Fluids	
Numerical investigation of Richtmyer-Meshkov instability in spherical geometry	2019
Wu, J., Liu, H. , Xiao, Z. Advances in Applied Mathematics and Mechanics	
Scale-to-scale energy transfer in mixing flow induced by the Richtmyer-Meshkov instability	2016
Liu, H. , Xiao, Z. Physical Review E	
Preferential concentration of heavy particles in compressible isotropic turbulence	2016
Zhang, Q., Liu, H. , Ma, Z., Xiao, Z. Physics of Fluids	

Selected Presentations

Enhancing incompressible two-phase turbulence simulations with quantum tensor networks	APS Global Summit 2025, Anaheim, CA 2025
Quantum-inspired high-fidelity solver for simulating incompressible two-phase turbulence	77th APS Division of Fluid Dynamics Annual Meeting, Salt Lake City 2024
Efficiency-enhanced high-fidelity simulation solver for incompressible two-phase flows and fluid-structure interactions	76th APS DFD Annual Meeting ing

2023

Simulation-based study on turbulent canopy flows and Monami phenomenon	75th APS DFD Annual Meeting, Indianapolis 2022
Simulation-based study on the interaction between flexible canopy and turbulent boundary layer flows	AGU 2022, Chicago 2022
Variational quantum algorithm towards quantum computing for fluid mechanics	74th APS DFD Annual Meeting, Phoenix 2021
Simulation-based study of airborne transmission of COVID-19 in two practical settings	73rd APS DFD Annual Meeting (Virtual) 2020
Simulation-based study of wind effect on shoaling waves	Ocean Sciences Meeting, San Diego 2020
A stabilized coupled level set and volume of fluid method	72nd APS DFD Annual Meeting, Seattle 2019
Numerical study on supercavitation turbulence	71st APS DFD Annual Meeting, Atlanta 2018
Simulation based study of supercavitation turbulence	10th International Workshop on Ship and Marine Hydrodynamics, Taiwan 2017
Numerical study on the interaction between supercavitation and turbulence	69th APS DFD Annual Meeting, Portland 2016
Preferential concentration of particles in compressible turbulence	15th European Turbulence Conference (ETC15), Delft, Netherlands 2015
Direct numerical simulation of the turbulent mixing in Richtmyer-Meshkov Instability	6th European Conference on Computational Fluid Dynamics (ECFD VI), Barcelona 2014
Energy transfer during the turbulent mixing in Richtmyer-Meshkov Instability	2nd International Conference on High Energy Density Physics (ICHEDP), Beijing 2014

Professional Memberships

- American Physical Society (APS)
- American Geophysical Union (AGU)

Peer Review Experience

- Physical Review Letters

- Journal of Fluid Mechanics
- Geophysical Research Letters
- Scientific Reports
- Physics of Fluids
- Review of Scientific Instruments
- Journal of Computational Science

Service

Session Chair: Computational Fluid Dynamics: General III

APS Division of Fluid Dynamics Annual Meeting
2021

Media Coverage

The Guardian: Bob Pape was a beloved father and foster carer. Did 'eat out to help out' cost him his life?

2021

- Research on COVID-19 airborne transmission featured in The Guardian